Sri Lanka Institute of Information Technology

B.Sc.Engineering (Hons) Degree

Mid-Semester Examination

Year 1 - Semester 2 (2025)

MT1011 - Engineering Materials

Duration: 1 Hour

March 2025

Instructions to candidates:

- This paper contains 25 multiple choice questions in 4 pages excluding the cover page and answer grid.
- Each question has four answer choices.
- Read each question carefully and choose the best answer OR phrase that completes the sentence.
- Tick the box corresponding to the most appropriate answer for questions in the answer grid provided.
- A reference sheet of A4 size, containing subject information and formulae, and hand written on both sides, is allowed to be brought into the examination hall as reference material.
- This is a closed-book examination.
- Detach the sheet containing the answer grid and submit the sheet along with the reference sheet.
- Faculty of engineering approved non-programmable calculators are allowed.
- This is an open book examination.

Index Number: EN .....

Answer Grid

	A	B	C	D
1	☐	☐	☐	☐
2	☐	☐	☐	☐
3	☐	☐	☐	☐
4	☐	☐	☐	☐
5	☐	☐	☐	☐
6	☐	☐	☐	☐
7	☐	☐	☐	☐
8	☐	☐	☐	☐
9	☐	☐	☐	☐
10	☐	☐	☐	☐
11	☐	☐	☐	☐
12	☐	☐	☐	☐
13	☐	☐	☐	☐
14	☐	☐	☐	☐
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21	☐	☐	☐	☐
22	☐	☐	☐	☐
23	☐	☐	☐	☐
24	☐	☐	☐	☐
25	☐	☐	☐	☐

Multiple choice questions **[100 marks]**

In the answer grid provided, tick the box corresponding to the most appropriate answer for questions 1–25. Each question has four answer choices. Read each question carefully and choose the best answer or the best phrase that completes the sentence. Each question carries 4 marks.

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1. Which of the following has the electron configuration $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^0\,3d^8$?

   (A) Ni (B) Fe **(C) Ni^{2+}** (D) Fe^{2+}

2. Letters A to E in Figure 1 illustrates schematically several types of defects find in crystals. Which of these can be classified as an substitutional defects?

   (A) *A* and *E* (B) *D* and *E* **(C) C and D** (D) *B* and *C*

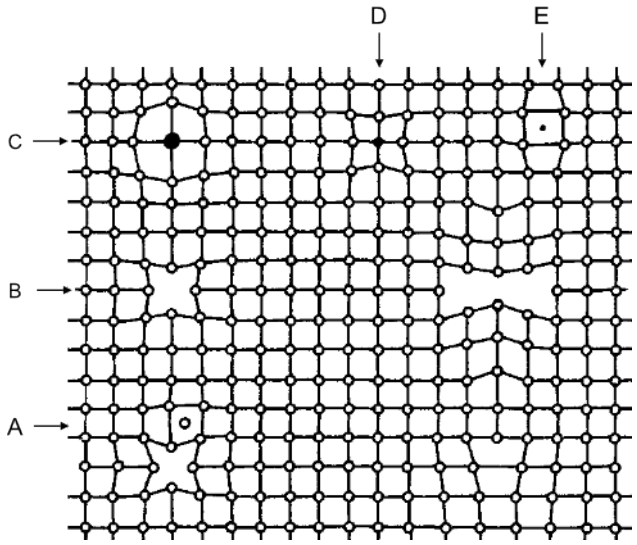


Figure 1

3. The percentage ionic character of the bonding in CuCl is approximately (use the electronegativity data given in Table 1):

   (A) 74.0 **(B) 26.1** (C) -35.3 (D) 99.8

Element	Cu	F	Cl	S	O
Electronegativity	1.9	4.0	3.0	2.5	3.5

Table 1: Electronegativities of elements.

4. Which of the following copper compound has bonding most covalent in nature?

   (A) CuF (B) CuCl **(C) CuS** (D) Cu_2O

5. Which plane is shown in Figure 2?

   (A) $(1\,0\,0)$ **(B)** $(0\,1\,0)$ (C) $(0\,0\,1)$ (D) $(0\,0\,\bar{1})$

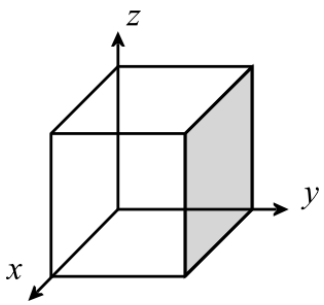


Figure 2

6. If the atomic radius is R, what is the areal density of atoms in the $(1\,1\,0)$ plane of the FCC crystal structure?

   (A) $\dfrac{3}{8\sqrt{2}R^2}$ (B) $\dfrac{3}{16\sqrt{2}R^2}$ **(C)** $\dfrac{1}{4\sqrt{2}R^2}$ (D) $\dfrac{1}{8R^2}$

7. What are the indices of the direction shown in Figure 3 for the cubic unit cell shown? (note that the arrow head is at the face centre).

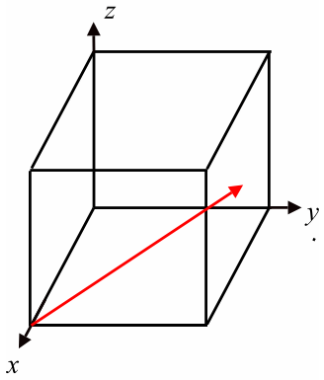


Figure 3

   (A) $[\bar{1}\,2\,1]$ (B) $[1\,1\,\bar{2}]$ (C) $[\bar{2}\,\bar{2}\,1]$ (D) $[\bar{2}\,\bar{1}\,2]$

8. Iron has an atomic radius of $0.124\,\mathrm{nm}$ and has an BCC crystal structure at room temperature. What is the linear density of iron atoms along the $[1\,1\,1]$ direction in atoms/nm?

   (A) 5.70 (B) 8.06 (C) 2.85 **(D) 4.03**

9. According to the gold-germanium (Au-Ge) phase diagram shown in Figure 4, what is the melting point of pure gold (Au)?

   (A) $1064.43\,^{\circ}\mathrm{C}$ (B) $938.3\,^{\circ}\mathrm{C}$ (C) $361.0\,^{\circ}\mathrm{C}$ (D) $862.0\,^{\circ}\mathrm{C}$

10. According to Figure 4, what is the eutectic temperature?

    (A) $1064.43\,^{\circ}\mathrm{C}$ (B) $938.3\,^{\circ}\mathrm{C}$ **(C)** $361.0\,^{\circ}\mathrm{C}$ (D) $862.0\,^{\circ}\mathrm{C}$

11. According to Figure 4, what is the eutectic composition?

    (A) 1 wt.% Au (B) 12.5 wt.% Au **(C) 12.5 wt.% Ge** (D) 1 wt.% Ge

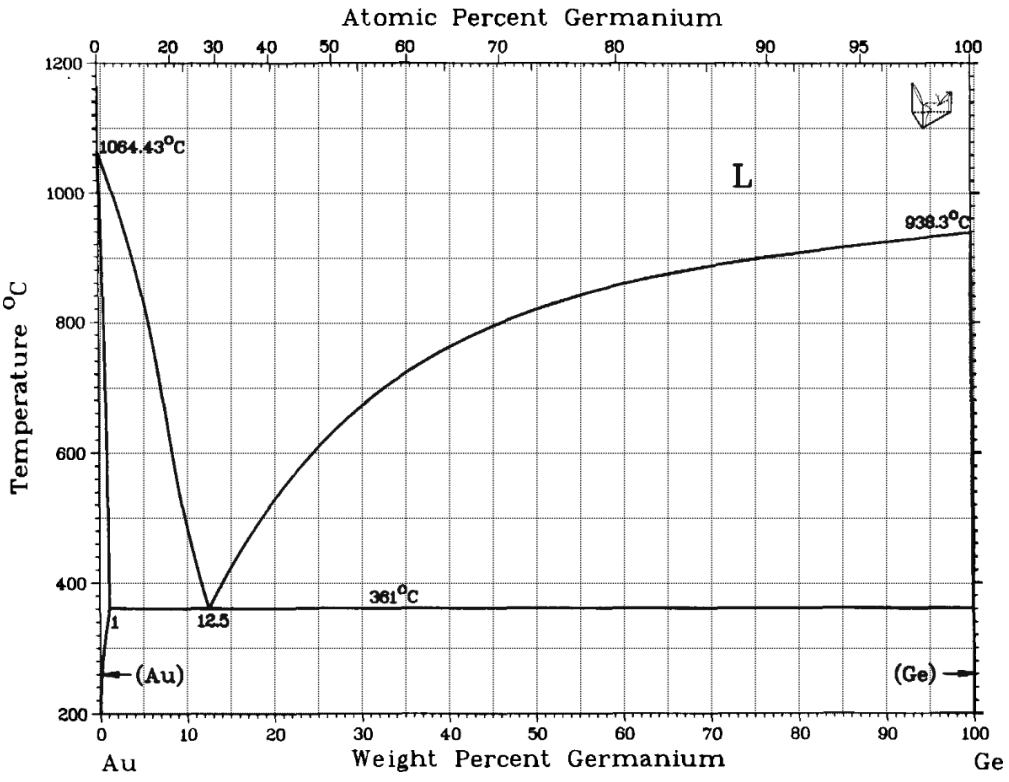


Figure 4

12. According to the Au-Ge phase diagram, if a liquid containing 75 wt.% Ge is cooled slowly from $1200\,^{\circ}\mathrm{C}$, the first solid will start to appear at

    (A) $1000\,^{\circ}\mathrm{C}$ (B) $950\,^{\circ}\mathrm{C}$ **(C)** $900\,^{\circ}\mathrm{C}$ (D) $850\,^{\circ}\mathrm{C}$

13. When the liquid containing 75 wt.% Ge is cooled slowly from $1200\,^{\circ}\mathrm{C}$, the solidification will be completed at

    (A) $938\,^{\circ}\mathrm{C}$ (B) $453\,^{\circ}\mathrm{C}$ **(C)** $361\,^{\circ}\mathrm{C}$ (D) $202\,^{\circ}\mathrm{C}$

14. During the cooling of the composition containing 75 wt.% the composition of the liquid at $700\,^{\circ}\mathrm{C}$ will be approximately

    (A) 32.5 wt.% Ge (B) 5.0 wt.% Ge (C) 100.0 wt.% Ge (D) 6.8 wt.% Ge

15. During the cooling of the composition containing 75 wt.% the liquid fraction at $800\,^{\circ}\mathrm{C}$ will be approximately

    (A) 0.46 (B) 0.54 (C) 0.27 (D) 0.73

16. After complete solidification, the microstructure of an alloy containing 75 wt.% Ge will consist of

    (A) Au-Ge eutectic **(B) Au-Ge eutectic and primary (pro-eutectic) Ge**
    (C) Au-Ge eutectic and primary (pro-eutectic) Au (D) Primary (pro-eutectic) Ge and Au

17. Which of the following belongs to the $\{111\}$ family of planes in the cubic system?

    (A) $(01\bar{1})$ (B) $(0\bar{1}0)$ **(C)** $(11\bar{1})$ (D) $(0\bar{1}1)$

18. Which of the following lattice points lie in the (110) plane of the FCC unit cell?

    (A) $1, 1, 1$ **(B)** $\frac{1}{2}, \frac{1}{2}, 1$ (C) $\frac{1}{2}, 0, \frac{1}{2}$ (D) $0, \frac{1}{2}, \frac{1}{2}$

19. If the atomic number of an element is Z and its atomic weight is A, the number of protons in the nucleus is

    (A) $A - Z$ (B) $Z - A$ (C) $A - Z - 2$ **(D)** Z

20. Which type of defect is responsible for slip and plastic deformation of crystals?

    (A) Grain boundaries (B) Vacancies **(C) Dislocations** (D) Interstitial atoms

21. Which of the following bond is the weakest?

    (A) Hydrogen (B) Metallic (C) Ionic (D) Covalent

22. A strong bond can be formed between a cloud of free moving valence electrons of sodium atoms is bonded to the positively charged ions of the sodium. It can be described as the sharing of free electrons among a lattice of positively charged sodium ions. This type of bonding is called

    (A) Metallic bond (B) Ionic bond (C) Covalent bond (D) Van der Walls bonds

23. Which of the following is NOT applicable to strengthening in the case of a pure metal?

    (A) Reducing the grain boundary area
    (B) Reducing the grain size
    (C) Work hardening
    (D) Solid solution strengthening

24. Atoms in a covalent bond is held together by

    (A) coulombic attraction **(B) sharing electrons** (C) gaining electrons (D) dipoles

25. What are the most common bonding types found in ceramics?

    (A) Ionic and metallic **(B) Covalent and ionic** (C) Covalent and hydrogen
    (D) Ionic and hydrogen

END OF QUESTION PAPER